```
> restart;
  #The Wave Equation Scale (where hbar noted as h because is
  more covenable);20 Feb.2026
```

```
> eq6 := 2 m / h^2 · ( E + ec^2/ρ − L^2/(2·m·ρ^2) − h^2/(2 ·m) k ) · ρ^2/R · f(ρ) + diff(ρ·f(ρ), ρ, ρ);
  #Onduscular;
```

$$eq6 := \frac{2\,m\left(E+\frac{ec^2}{\rho}-\frac{L^2}{2\,m\,\rho^2}-\frac{h^2\,k}{2\,m}\right)\rho^2 f(\rho)}{h^2\,R}+2\,\frac{\mathrm{d}}{\mathrm{d}\rho}f(\rho)+\rho\left(\frac{\mathrm{d}^2}{\mathrm{d}\rho^2}f(\rho)\right) \tag{1}$$

```
> change_of_vars := { rho = rb·r, f(rho) = g(r)};
```

$$change_of_vars := \{\rho = rb\,r,\ f(\rho) = g(r)\} \tag{2}$$

```
> PDEtools:-dchange(change_of_vars, eq6, [g(r), r]);
```

$$\frac{2\,m\left(E+\frac{ec^2}{rb\,r}-\frac{L^2}{2\,m\,rb^2\,r^2}-\frac{h^2\,k}{2\,m}\right)rb^2\,r^2\,g(r)}{h^2\,R}+\frac{2\left(\frac{\mathrm{d}}{\mathrm{d}r}g(r)\right)}{rb}+\frac{r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}g(r)\right)}{rb} \tag{3}$$

```
> eq1 := 2 m ( E + ec^2/(rb r) − L^2/(2 m rb^2 r^2) − h^2 k/(2 m) ) rb^2 r^2 g(r) / (h^2 R)
    + 2 (d/dr g(r))/rb + r (d^2/dr^2 g(r))/rb;
```

$$eq1 := \frac{2\,m\left(E+\frac{ec^2}{rb\,r}-\frac{L^2}{2\,m\,rb^2\,r^2}-\frac{h^2\,k}{2\,m}\right)rb^2\,r^2\,g(r)}{h^2\,R}+\frac{2\left(\frac{\mathrm{d}}{\mathrm{d}r}g(r)\right)}{rb}+\frac{r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}g(r)\right)}{rb} \tag{4}$$

```
> subs( E = m·ec^4/(32·Pi^2·ϵ^2·h^2)·E0, rb = 4·Pi·ϵ· h^2/(m·ec^2), eq1 );
```

$$\frac{32\,h^2\left(\frac{m\,ec^4\,E0}{32\,\pi^2\,\epsilon^2\,h^2}+\frac{ec^4\,m}{4\,\pi\,\epsilon\,h^2\,r}-\frac{L^2\,m\,ec^4}{32\,\pi^2\,\epsilon^2\,h^4\,r^2}-\frac{h^2\,k}{2\,m}\right)\pi^2\,\epsilon^2\,r^2\,g(r)}{m\,ec^4\,R}+\frac{\left(\frac{\mathrm{d}}{\mathrm{d}r}g(r)\right)m\,ec^2}{2\,\pi\,\epsilon\,h^2} \tag{5}$$

$$+\frac{m\,ec^2\,r\left(\frac{d^2}{dr^2}\,g(r)\right)}{4\,\pi\,\epsilon\,h^2}$$

```
> simplify(%);
```

$$\frac{1}{4\,h^2\,\pi\,\epsilon\,m^2\,ec^4\,R}\left(m^3\,ec^6\,r\left(\frac{d^2}{dr^2}\,g(r)\right)R+2\left(\frac{d}{dr}\,g(r)\right)m^3\,ec^6\,R-4\,\pi\,\epsilon\left(m^2\left(-E0\,h^2\,r^2-8\,\pi\,\epsilon\,h^2\,r+L^2\right)ec^4+16\,\epsilon^2\,h^6\,k\,r^2\,\pi^2\right)g(r)\right) \tag{6}$$

```
> r*(diff(g(r), r, r))+2*diff(g(r), r)-4*g(r)*(m^2*(-E0*h^2*r^2 -
  8*Pi*epsilon*h^2*r + L^2)*ec^4 + 16*epsilon^2*h^6*k*r^2*Pi^2)*
  epsilon*Pi*m*ec^2/(4*h^2*Pi);
```

$$r\left(\frac{d^2}{dr^2}\,g(r)\right)+2\,\frac{d}{dr}\,g(r)-\frac{g(r)\left(m^2\left(-E0\,h^2\,r^2-8\,\pi\,\epsilon\,h^2\,r+L^2\right)ec^4+16\,\epsilon^2\,h^6\,k\,r^2\,\pi^2\right)\epsilon\,m\,ec^2}{h^2} \tag{7}$$

```
> simplify((-m^2*(-E0*h^2*r^2 - 8*Pi*epsilon*h^2*r + L^2)*ec^4 +
  16*epsilon^2*h^6*k*r^2*Pi^2)*epsilon*m*ec^2/h^2);
```

$$-\frac{ec^2\left(\left(\left(-E0\,r^2-8\,\pi\,\epsilon\,r\right)h^2+L^2\right)m^2\,ec^4-16\,\epsilon^2\,h^6\,k\,r^2\,\pi^2\right)\epsilon\,m}{h^2} \tag{8}$$

```
> eq2 := (E0 + 8·Pi·ε/r - L^2/(h^2·r^2) - 16·h^4·k·π^2·ε^2/(m^2·ec^4))·r^2/R·g(r) + diff(r·g(r), r, r);
```

$$eq2 := \frac{\left(E0+\frac{8\,\pi\,\epsilon}{r}-\frac{L^2}{h^2\,r^2}-\frac{16\,h^4\,k\,\pi^2\,\epsilon^2}{m^2\,ec^4}\right)r^2\,g(r)}{R}+2\,\frac{d}{dr}\,g(r)+r\left(\frac{d^2}{dr^2}\,g(r)\right) \tag{9}$$

```
>
  eq2 := (E0 + 8 π ε/r - L^2/(h^2·r^2) - 16 h^4 k π^2 ε^2/(m^2 ec^4))·r^2/R ·g(r) + 2 d/dr g(r) + r (d^2/dr^2
  g(r));
```

$$eq2 := \frac{\left(E0+\frac{8\,\pi\,\epsilon}{r}-\frac{L^2}{h^2\,r^2}-\frac{16\,h^4\,k\,\pi^2\,\epsilon^2}{m^2\,ec^4}\right)r^2\,g(r)}{R}+2\,\frac{d}{dr}\,g(r)+r\left(\frac{d^2}{dr^2}\,g(r)\right) \tag{10}$$

```
> TFreeOrd := (E0/R + 8*π*ϵ/(r·R) - L^2/(h^2*r^2·R) - 16*h^4*k*π^2*ϵ^2/(m^2*ec^4·R))·r^2·g(r);
```

$$TFreeOrd := \left(\frac{E0}{R} + \frac{8\,\pi\,\epsilon}{r\,R} - \frac{L^2}{h^2\,r^2\,R} - \frac{16\,h^4\,k\,\pi^2\,\epsilon^2}{m^2\,ec^4\,R}\right) r^2\,g(r) \tag{11}$$

```
> sort(((E0/R + 8*π*ϵ/(r*R) - L^2/(h^2*r^2*R) - 16*h^4*k*π^2*ϵ^2/(m^2*ec^4*R))*r^2), r);
```

$$\left(\frac{8\,\pi\,\epsilon}{R\,r} - \frac{L^2}{h^2\,R\,r^2} + \frac{E0}{R} - \frac{16\,h^4\,k\,\pi^2\,\epsilon^2}{m^2\,ec^4\,R}\right) r^2 \tag{12}$$

```
> eq2 := (E0*r^2/R - 16*h^4*k*π^2*ϵ^2*r^2/(m^2*ec^4*R) + 8*π*ϵ*r/R - L^2/(h^2*R))·g(r) + 2*d/dr g(r)
      + r*(d^2/dr^2 g(r));#Onduscular equation.
```

$$eq2 := \left(\frac{E0\,r^2}{R} - \frac{16\,h^4\,k\,\pi^2\,\epsilon^2\,r^2}{m^2\,ec^4\,R} + \frac{8\,\pi\,\epsilon\,r}{R} - \frac{L^2}{h^2\,R}\right) g(r) + 2\,\frac{\mathrm{d}}{\mathrm{d}r}\,g(r) + r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right) \tag{13}$$

```
> restart;
    #ScrodingeScal where hbar noted as h because is more
    covenable;we start from eqation 19.8 Feyman
```

```
> eq1 := 2*m/h^2·(E + ec^2/ρ)·ρ·f(ρ) + diff(ρ·f(ρ), ρ, ρ);#Schrodinger;
```

$$eq1 := \frac{2\,m\left(E + \frac{ec^2}{\rho}\right)\rho\,f(\rho)}{h^2} + 2\,\frac{\mathrm{d}}{\mathrm{d}\rho}\,f(\rho) + \rho\left(\frac{\mathrm{d}^2}{\mathrm{d}\rho^2}\,f(\rho)\right) \tag{14}$$

```
> change_of_vars := { rho = rb·r, f(rho) = g(r)};
```

$$change_of_vars := \{\rho = rb\,r, f(\rho) = g(r)\} \tag{15}$$

```
> PDEtools:-dchange(change_of_vars, eq1, [g(r), r]);
```

$$\frac{2\,m\left(E + \frac{ec^2}{rb\,r}\right) rb\,r\,g(r)}{h^2} + \frac{2\left(\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)\right)}{rb} + \frac{r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right)}{rb} \tag{16}$$

```
> eq2 := 2*m*(E + ec^2/(rb*r))*rb^2*r*g(r)/h^2 + d^2/dr^2(r·g(r));#we multiply rb ;
```

$$eq2 := \frac{2\,m\left(E+\frac{ec^2}{rb\,r}\right)rb^2\,r\,g(r)}{h^2}+2\,\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)+r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right) \tag{17}$$

```
> subs(E = m·ec^4/(32·Pi^2·ε^2·h^2)·E0, rb = 4·Pi·ε·h^2/(m·ec^2), eq2);
  #with wacuum permittivity ε and E0=energy in Rythberg unit
```

$$\frac{32\left(\frac{m\,ec^4\,E0}{32\,\pi^2\,\epsilon^2\,h^2}+\frac{ec^4\,m}{4\,\pi\,\epsilon\,h^2\,r}\right)\pi^2\,\epsilon^2\,h^2\,r\,g(r)}{m\,ec^4}+2\,\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)+r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right) \tag{18}$$

```
> eq2 := 32 ·(E0/(32· h^2)+Pi·ε/(4· h^2 ·r)) ·h^2 ·r ·g(r)+2 d/dr g(r)+r (d^2/dr^2 g(r));
```

$$eq2 := 32\left(\frac{E0}{32\,h^2}+\frac{\pi\,\epsilon}{4\,h^2\,r}\right)h^2\,r\,g(r)+2\,\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)+r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right) \tag{19}$$

```
> eq2 := 32 (E0/32+π ε/(4· r))· r ·g(r)+2 d/dr g(r)+r (d^2/dr^2 g(r));
```

$$eq2 := 32\left(\frac{E0}{32}+\frac{\pi\,\epsilon}{4\,r}\right)r\,g(r)+2\,\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)+r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right) \tag{20}$$

```
> eq2 := (E0+8·π·ε/r)· r ·g(r)+d^2/dr^2( r·g(r));
```

$$eq2 := \left(E0+\frac{8\,\pi\,\epsilon}{r}\right)r\,g(r)+2\,\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)+r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right) \tag{21}$$

```
> restart;
  # we have the Schrodinger Scaled equation with wacuum
  permittivity ε
> eq3 := (E0+8·π·ε/ρ)·ρ·f(ρ)+d^2/dρ^2(ρ·f(ρ));
  #eqation (1) with wacuum permittivity ε
```

$$eq3 := \left(E0+\frac{8\,\pi\,\epsilon}{\rho}\right)\rho\,f(\rho)+2\,\frac{\mathrm{d}}{\mathrm{d}\rho}\,f(\rho)+\rho\left(\frac{\mathrm{d}^2}{\mathrm{d}\rho^2}\,f(\rho)\right) \tag{22}$$

```
> restart;
  #now we have the Schrodinger Scaled eq1 with wacuum
  permittivity ε;
```

```
> eq1 := (E0 + 8·π·ϵ/ρ)·  ρ ·f(ρ) + d^2/dρ^2 (ρ·f(ρ));
```

$$eq1 := \left(E0 + \frac{8\,\pi\,\epsilon}{\rho}\right)\rho f(\rho) + 2\,\frac{d}{d\rho}f(\rho) + \rho\left(\frac{d^2}{d\rho^2}f(\rho)\right) \tag{23}$$

```
> change_of_vars := { rho = rb·r, f(rho) = g(r)·(R − r·rb)/R };
```

$$change_of_vars := \left\{\rho = rb\,r, f(\rho) = \frac{g(r)\,(-rb\,r+R)}{R}\right\} \tag{24}$$

```
> PDEtools:-dchange(change_of_vars, eq1, [g(r), r]);
```

$$\frac{\left(E0 + \frac{8\,\pi\,\epsilon}{rb\,r}\right) rb\,r\,g(r)\,(-rb\,r+R)}{R} + \frac{2\left(\frac{\left(\frac{d}{dr}g(r)\right)(-rb\,r+R)}{R} - \frac{g(r)\,rb}{R}\right)}{rb} + \frac{r\left(\frac{\left(\frac{d^2}{dr^2}g(r)\right)(-rb\,r+R)}{R} - \frac{2\left(\frac{d}{dr}g(r)\right)rb}{R}\right)}{rb} \tag{25}$$

```
> eq2 := simplify(%);
```

$$eq2 := \frac{1}{rb\,R}\left(r\,(-rb\,r+R)\left(\frac{d^2}{dr^2}g(r)\right) + (-4\,rb\,r+2\,R)\left(\frac{d}{dr}g(r)\right) + g(r)\,(-r^2\,E0\,rb^2 + r\,(E0\,R - 8\,\pi\,\epsilon)\,rb + 8\,R\,\pi\,\epsilon - 2)\,rb\right) \tag{26}$$

```
> eq2 := %/(R − r·rb);#we devide with (−r rb+ R );
```

$$eq2 := \frac{1}{rb\,R\,(-rb\,r+R)}\left(r\,(-rb\,r+R)\left(\frac{d^2}{dr^2}g(r)\right) + (-4\,rb\,r+2\,R)\left(\frac{d}{dr}g(r)\right) + g(r)\,(-r^2\,E0\,rb^2 + r\,(E0\,R - 8\,\pi\,\epsilon)\,rb + 8\,R\,\pi\,\epsilon - 2)\,rb\right) \tag{27}$$

```
>
```

```
> eq2 := r/(rb R)·(d^2/dr^2 g(r)) + (−4 rb r + 2·R)/(rb R·(−r rb + R))·(d/dr g(r)) + g(r)/(R·(−r rb + R))
      ·(−r^2 E0 rb^2 + r (E0 R − 8 π ϵ) rb + 8 R π ϵ − 2);
      #We aproximate (−4 rb r + 4 R)/(−r rb + 2 R) ~ 2;
```

$$eq2 := \frac{r\left(\frac{d^2}{dr^2}\,g(r)\right)}{rb\,R} + \frac{(-4\,rb\,r + 2\,R)\left(\frac{d}{dr}\,g(r)\right)}{rb\,R\,(-rb\,r + R)} + \frac{g(r)\,\left(-r^2\,E0\,rb^2 + r\,(E0\,R - 8\,\pi\,\epsilon)\,rb + 8\,R\,\pi\,\epsilon - 2\right)}{R\,(-rb\,r + R)} \tag{28}$$

```
> eq2 := r*(diff(g(r),r$2)) + 2*diff(g(r),r)
       + g(r)*rb*(-r^2*E0*rb^2 + r*(E0*R + 8*Pi*eps)*rb + 8*R*Pi*eps - 2)/(-rb*r + R);
  #We multiply with rb·R
```

$$eq2 := r\left(\frac{d^2}{dr^2}\,g(r)\right) + 2\,\frac{d}{dr}\,g(r) + \frac{g(r)\,rb\,\left(-r^2\,E0\,rb^2 + r\,(E0\,R + 8\,\pi\,\epsilon)\,rb + 8\,R\,\pi\,\epsilon - 2\right)}{-rb\,r + R} \tag{29}$$

```
> TFreeOrd :=
      simplify(rb*(-r^2*E0*rb^2 + r*(E0*R + 8*Pi*eps)*rb + R*Pi*eps - 2)/(-rb*r + R));
```

$$TFreeOrd := \frac{rb\,\left(-r^2\,E0\,rb^2 + r\,(E0\,R + 8\,\pi\,\epsilon)\,rb + R\,\pi\,\epsilon - 2\right)}{-rb\,r + R} \tag{30}$$

```
> series(rb*(-r^2*E0*rb^2 + r*(E0*R + 8*Pi*eps)*rb + 8*R*Pi*eps - 2)/(-rb*r + R), r);
```

$$\frac{rb\,(8\,R\,\pi\,\epsilon - 2)}{R} + \frac{(E0\,R + 8\,\pi\,\epsilon)\,rb^2 + \frac{2\,rb^2\,(4\,R\,\pi\,\epsilon - 1)}{R}}{R}\,r + \frac{-rb^3\,E0 + \frac{rb^3\,(E0\,R^2 + 16\,R\,\pi\,\epsilon - 2)}{R^2}}{R}\,r^2 + 2\,\frac{rb^4\,(8\,R\,\pi\,\epsilon - 1)}{R^4}\,r^3 + 2\,\frac{rb^5\,(8\,R\,\pi\,\epsilon - 1)}{R^5}\,r^4 + 2\,\frac{rb^6\,(8\,R\,\pi\,\epsilon - 1)}{R^6}\,r^5 + O\left(r^6\right) \tag{31}$$

```
> eq3 := r*(diff(g(r),r$2)) + 2*diff(g(r),r) + (rb*(+8*R*Pi*eps - 2)/R + rb
```

$$\cdot \frac{\left(E0\ R+8\ \pi\ \epsilon\right)\ rb^2-\frac{2\ \left(4\ R\ \pi\ \epsilon+1\right)\ rb^2}{R}}{R}\ r$$

$$+\frac{-rb^3\ E0+\frac{rb^3\ \left(E0\ R^2-2\right)}{R^2}}{R}\ r^2\Bigg)\cdot g(r);\#Aproximative$$

$$eq3 := r\left(\frac{d^2}{dr^2}\ g(r)\right)+2\ \frac{d}{dr}\ g(r)+\left(\frac{rb\left(8\ R\ \pi\ \epsilon-2\right)}{R}\right. \tag{32}$$

$$+\frac{rb\left(\left(E0\ R+8\ \pi\ \epsilon\right)\ rb^2-\frac{2\left(4\ R\ \pi\ \epsilon+1\right)\ rb^2}{R}\right)r}{R}$$

$$\left.+\frac{\left(-rb^3\ E0+\frac{rb^3\left(E0\ R^2-2\right)}{R^2}\right)r^2}{R}\right)g(r)$$

$$> rb\cdot\frac{+8\ R\ \pi\ \epsilon-2}{R}+\frac{rb\left(\left(E0\ R+8\ \pi\ \epsilon\right)\ rb^2-\frac{2\ \left(4\ R\ \pi\ \epsilon+1\right)\ rb^2}{R}\right)}{R}\ r$$

$$+\frac{\left(-rb^3\ E0+\frac{rb^3\ \left(E0\ R^2-2\right)}{R^2}\right)}{R}\ r^2=\frac{E0\ r^2}{R}-\frac{16\ h^4\ k\ \pi^2\ \epsilon^2\ r^2}{m^2\ ec^4\ R}-\frac{8\ \pi\ \epsilon\ r}{R}$$

$$+\frac{L^2}{h^2\ R};\#Aproximation;$$

$$\frac{rb\left(8\ R\ \pi\ \epsilon-2\right)}{R}+\frac{rb\left(\left(E0\ R+8\ \pi\ \epsilon\right)\ rb^2-\frac{2\left(4\ R\ \pi\ \epsilon+1\right)\ rb^2}{R}\right)r}{R} \tag{33}$$

$$+\frac{\left(-rb^3\ E0+\frac{rb^3\left(E0\ R^2-2\right)}{R^2}\right)r^2}{R}=\frac{E0\ r^2}{R}-\frac{16\ h^4\ k\ \pi^2\ \epsilon^2\ r^2}{m^2\ ec^4\ R}-\frac{8\ \pi\ \epsilon\ r}{R}+\frac{L^2}{h^2\ R}$$

$$> solve\left(\left\{\frac{\left(-rb^3\ E0+\frac{rb^3\ \left(E0\ R^2-2\right)}{R^2}\right)}{R}=\frac{E0}{R}-\frac{16\ h^4\ k\ \pi^2\ \epsilon^2}{m^2\ ec^4\ R},\right.$$

$$\left(\frac{\left(rb\left(2\ E0\ R\ rb-8\ \pi\ \epsilon\ rb\right)+\frac{rb^2\left(8\ R\ \pi\ \epsilon-1\right)}{R}\right)}{2\ R}=-\frac{8\ \pi\ \epsilon}{R},\right.$$

$$\left.\frac{rb\left(8\ R\ \pi\ \epsilon-2\right)}{R}=-\frac{L^2}{h^2\ R}\right\},\ \{R,\ k,\ E0\}\Bigg);$$

$$\left\{E0=\frac{32\,\pi^2\,\epsilon^2\,h^2\left(h^2\,rb^3-4\,rb\,h^2+2\,L^2\right)}{rb\left(4\,h^4\,rb^2-4\,L^2\,h^2\,rb+L^4\right)},\ R=-\frac{-2\,rb\,h^2+L^2}{8\,\pi\,\epsilon\,h^2\,rb},\ k\right.$$

$$\left.=\frac{2\,m^2\,ec^4\left(4\,h^2\,rb^6+h^2\,rb^3-4\,rb\,h^2+2\,L^2\right)}{rb\,h^2\left(4\,h^4\,rb^2-4\,L^2\,h^2\,rb+L^4\right)}\right\} \tag{34}$$

> $R:=\frac{2\ rb\ h^2-L^2}{8\ \pi\ \epsilon\ h^2\ rb};$

$$R:=\frac{2\,rb\,h^2-L^2}{8\,\pi\,\epsilon\,h^2\,rb} \tag{35}$$

> $subs\left(rb=\frac{4\cdot Pi\cdot\epsilon\cdot\ h^2}{m\cdot ec^2},\ \%\right);$

$$\frac{\left(\frac{8\,\pi\,\epsilon\,h^4}{m\,ec^2}-L^2\right)m\,ec^2}{32\,\pi^2\,\epsilon^2\,h^4} \tag{36}$$

> $simplify(\%);\#The\ R\ value$

$$\frac{8\,\pi\,\epsilon\,h^4-L^2\,m\,ec^2}{32\,\pi^2\,\epsilon^2\,h^4} \tag{37}$$

> **#We substitute according convetion h with hbar**

> **m := 9.10938356*10^(-31); epsilon := 8.8541878128*10^(-12);hbar := 6.62607015/(2*Pi)*10^(-34); ec := 1.602176*10^(-19); c := 299792458; h := 6.62607015*10^(-34); rb := 5.292410932*10^(-11);L:=(21.84*22.84)^0.5*hbar;metru := 1.057023410*10^(-16);l:=21.84;**

$$m := 9.109383560 \times 10^{-31}$$

$$\epsilon := 8.854187813 \times 10^{-12}$$

$$\hbar := 1.054571818 \times 10^{-34}$$

$$ec := 1.602176000 \times 10^{-19}$$

$$c := 299792458$$

$$h := 6.626070150 \times 10^{-34}$$

$$rb := 5.292410932 \times 10^{-11}$$

$$L := 2.355323298 \times 10^{-33}$$

$$metru := 1.057023410 \times 10^{-16}$$

$$l := 21.84 \quad \textbf{(38)}$$

```
> R:=-(8*Pi*epsilon*hbar^4 - L^2*ec^2*m)/(32*Pi^2*epsilon^2*hbar^4)
  ;R*metru;R := (-hbar^2*rb + L^2)/(8*Pi*epsilon*hbar^2*rb);# The R
  value in m and in LightYears;
```

$$R := 4.236026422 \times 10^{22}$$

$$4.477579093 \times 10^{6}$$

$$R := 4.235518534 \times 10^{22} \quad \textbf{(39)}$$

```
> R:=l*(l+1)/(8*Pi*epsilon*rb);R*metru;
```

$$R := 4.235518529 \times 10^{22}$$

$$4.477042239 \times 10^{6} \quad \textbf{(40)}$$

```
> L:=(21.84*22.84)^0.5*hbar;(-hbar^2*rb + L^2)/(8*Pi*epsilon*hbar^
  2*rb);%*metru;
```

$$L := 2.355323298 \times 10^{-33}$$

$$4.235518534 \times 10^{22}$$

$$4.477042244 \times 10^{6} \quad \textbf{(41)}$$

```
> (-hbar^2*rb + L^2)/(8*Pi*epsilon*hbar^2*rb)
```

$$4.235518534 \times 10^{22} \quad \textbf{(42)}$$

```
> l*(l+1)/(8*rb*Pi*epsilon);
```

$$4.235518529 \times 10^{22} \quad \textbf{(43)}$$

```
> l:= 21.84 ; R:=l*(l+1)/(8*Pi*epsilon*rb)
```

$$l := 21.84$$

$$R := 4.235518529 \times 10^{22} \quad \textbf{(44)}$$